The Reproducibility Crisis in Artificial Intelligence:

Statistical Biases, Benchmark Limitations, and Methodological Solutions

February 28, 2026

Abstract

Empirical research in Artificial Intelligence (AI) depends heavily on data-driven experiments, benchmarks, and complex software–hardware ecosystems. This dependence creates distinctive risks for reproducibility and reliable inference that fall into three broad categories: *(i)* statistical biases—including p-hacking analogues that arise from the many degrees of freedom in model and training decisions, as well as selection and publication biases; *(ii)* systematic over-adaptation to benchmarks, encompassing adaptive misuse of holdout sets and Goodhart effects; and *(iii)* methodological error classes such as overfitting, dataset leakage, duplicate samples, and label errors, all of which can substantially distort reported performance figures.

Meta-research and community programmes have documented persistent shortcomings in the reporting of code, data, and experimental configurations for years, yet they also demonstrate that institutional interventions—checklists, artifact reviews, reproducibility challenges—can trigger measurable behavioural change [1, 2].

This paper synthesises the scientific literature on the "reproducibility crisis" in AI along statistical, infrastructural, and governance-related dimensions [3, 4]. Furthermore, it presents a solution framework that combines Open Science practices, technical infrastructure (versioning, provenance, packaging), dataset governance (documentation standards), and automated citation and evidence verification to raise the validity of empirical AI results and to reduce the perverse incentives of the benchmark ecosystem.

Keywords: reproducibility, replication, p-hacking, publication bias, dataset leakage, benchmarking, adaptive data analysis, Open Science, data and software citation, provenance, versioning, evidence and citation verification.

Contents

1 Introduction **3**

2 Statistical and Methodological Causes **4**

2.1 Statistical Foundations: p-Hacking Analogues, Multiple Degrees of Freedom, and Publication Bias . . . 4

2.2 Overfitting and Dataset Leakage: Forms, Causes, and Empirical Evidence . 5

3 Limitations of Benchmarks and Optimisation Dynamics **6**

3.1 Benchmark Optimisation as an Adaptive Inference Problem . . . 6

3.2 Goodhart Effects, "General" Benchmarks, and External Validity . . . 7

3.3 Benchmark Design Beyond Leaderboards: Suites Instead of Single-Number Rankings . . . 7

4 Evidence from Meta-Research and Open Science Initiatives **8**

4.1 Meta-Research: Findings on Documentation and Reproduction Practice in AI 8

4.2 Open Science Initiatives: Checklists, Artifact Reviews, and Community Programmes . . . 9

5 Technical and Organisational Solutions **9**

5.1 Versioning, Provenance, and Executable Reproduction . . . 10

5.2 Dataset Governance: Documentation, Quality Assurance, and Citation Practices . . . 10

5.3 Automated Citation and Evidence Verification Systems . . . 11

6 Case Studies **12**

6.1 Reinforcement Learning: Variance, Non-Determinism, and Inference Uncertainty . . . 12

6.2 Computer Vision Benchmarks: Near-Duplicates and Measurement Foundations . . . 12

6.3 LLM Evaluation: Contamination and Dynamic Benchmarks . . . 12

6.4 ML-Based Science: Leakage as a Reproduction Error . . . 13

7 Discussion **13**

8 Conclusion **14**

1 Introduction

Reproducibility and replication are cornerstones of cumulative science, yet in practice they are compromised by statistical, social, and infrastructural factors across many disciplines. The very terminology remains incompletely standardised; a key finding of meta-science is that precise conceptual work—for instance, distinguishing *methods reproducibility*, *results reproducibility*, and *inferential reproducibility*—is a prerequisite for operationalising crisis diagnoses and evaluating interventions at all [3].

In AI—including machine learning (ML), deep learning, and data-intensive subfields such as natural language processing (NLP), computer vision, and reinforcement learning (RL)—the classical reproducibility risks are amplified by three structural properties of the field:

1. **Stochastic training and fragile baselines.** Central empirical claims frequently rest on small performance differences on benchmarks, yet these differences are subject to substantial fluctuation from stochastic training processes, non-deterministic execution environments, and hyperparameter search [5, 6].

2. **Complex, under-documented artefacts.** Research artefacts—code, data, models, preprocessing pipelines, dependencies, hardware and driver versions—are highly complex and often insufficiently documented or archived, impeding both repetition and independent verification [7–9].

3. **Benchmark-driven incentives and Goodhart dynamics.** Benchmarks and leaderboards create powerful incentive structures. When a metric becomes a target, it can lose its validity as a measurement instrument (Goodhart effect), particularly through adaptive overuse of static test sets and through metrics that lend themselves to gaming [10, 11].

The scientific community has responded over the past several years with programmatic and infrastructural measures: conferences have implemented checklists and code policies; artifact review processes and reproducibility challenges create new forms of publication and recognition; Open Science and FAIR principles target the findability, reusability, and machine-actionability of research objects [2, 12, 13]. At the same time, the literature shows that technical solutions (versioning, provenance, packaging, archivability) have limited effectiveness without adequate governance and without statistical methodological discipline, because biases can already arise in the research logic itself—in hypothesis generation, model selection, and reporting and publication dynamics [14, 15].

The goal of this paper is a deep, source-based synthesis of the reproducibility problem in AI along the following dimensions: statistical foundations (p-hacking, publication bias), overfitting and dataset leakage, benchmark optimisation problems, meta-research evidence,

Open Science initiatives, technical solutions (versioning, dataset governance, automated citation verification), case studies, and a concluding discussion that integrates these threads into a unified solution framework.

2 Statistical and Methodological Causes

Before examining AI-specific pathologies, it is useful to establish the broader statistical context in which the reproducibility crisis is rooted. Two mutually reinforcing mechanisms—flexible analysis decisions and selective publication—generate systematic distortions that inflate the apparent strength and reliability of empirical findings. This section traces these mechanisms from their general-science origins into their AI-specific manifestations, and then turns to the methodological error classes (overfitting, leakage, contamination) that amplify the problem.

2.1 Statistical Foundations: p-Hacking Analogues, Multiple Degrees of Freedom, and Publication Bias

The classical p-hacking debate demonstrates that flexible analysis decisions ("researcher degrees of freedom") and selective reporting can render significant results more probable even in the absence of robust effects. This phenomenon is well documented across many empirical sciences and is discussed as a central mechanism behind false-positive findings [16–18].

In AI, a functional analogue manifests not as manipulation of individual p-values but as optimisation across many degrees of freedom: choice of architecture, data augmentation, preprocessing, optimiser, training budget, random seed, hyperparameter space, and selection criterion (e.g., best single run versus mean across seeds). When only the best configuration is subsequently reported, a selection bias emerges that is formally equivalent to the multiple-testing problem [5, 6].

Several studies emphasise that pure "determinism" reproducibility (same code, same seed, identical environment) is insufficient to secure inferential robustness. Instead, variance from multiple sources—data sampling, initialisation, hyperparameter choice, implementation details—must be modelled and integrated into the interpretation of results [6, 19]. This demand is closely related to statistical guidelines for correct algorithm-comparison inference, such as appropriate tests over multiple datasets or experimental conditions [20, 21].

Publication bias (the "file drawer problem") further compounds the distortion: when "positive" or "state-of-the-art" results enjoy a higher publication probability than null or negative results, effect sizes and progress rates are systematically overestimated. This is well established in meta-analyses and population-based studies of publication dynamics

[22, 23]. For AI, the analogy is discussed primarily as an incentive problem: benchmarks, competitions, and community norms frequently reward incremental performance gains, whereas replication studies, negative results, or "ablation null findings" receive lesser recognition. Open Science manifestos and recommendations for reforming incentive and publication systems address these dynamics explicitly [13, 15].

Methodological counter-reactions from the philosophy of science and meta-science literature include, among others, preregistration, Registered Reports (review before data analysis or before knowledge of results), and stronger transparency requirements. Registered Reports have been established as a publication format to reduce results-orientation and to tie the publication decision more closely to methodology than to significance [24]. In AI, direct transfer is not trivial (e.g., iterative development, high experimental costs), yet the core principle—separating the analysis plan from the results—remains conceptually highly relevant as a countermeasure against selective reporting [3].

2.2 Overfitting and Dataset Leakage: Forms, Causes, and Empirical Evidence

The general statistical biases described above are compounded by a set of methodological error classes specific to the data-centric nature of ML research. Overfitting in AI is not merely a theoretical concept of the learning curve; it is tightly coupled with benchmark culture, data-pipeline design, and evaluation practices.

A central error class is **dataset leakage**: information from test or validation data enters training or tuning processes, whether implicitly or explicitly—for instance through incorrect splitting, preprocessing on the full dataset, leakage via temporal indices, or meta-feature engineering [14]. A comprehensive, cross-domain study argues that leakage in ML-based science is widespread and has in numerous cases led to "wildly overoptimistic conclusions"; it further proposes standardised "Model Info Sheets" to systematically audit leakage risks [14].

In domains with highly structured data (e.g., medical imaging), even the *granularity* of the split—patient/study/volume level versus image/slice level—can determine whether seemingly outstanding performance is real or artificially inflated by leakage. A detailed analysis of optical coherence tomography (OCT) image classification shows that inappropriate splits (with overlap between training and test sets) can markedly inflate accuracy figures, reporting large performance differences between correct and incorrect splitting strategies [25].

Alongside "classical" leakage, a second, structurally related problem class exists: **duplicates and near-duplicates** between training and test sets. For heavily used image datasets, it has been shown that a non-trivial proportion of test examples have duplicates in the training set; this can distort evaluation comparability because models effectively

(partially) train on test data or benefit from memorisation. A study on CIFAR-10/100 identifies near-duplicates in the test set, replaces them with fresh examples ("ciFAIR"), and observes significant accuracy drops—a direct indication that benchmark values can be systematically influenced by data overlap [26].

A third problem source consists of **label errors in test sets**. A systematic characterisation of label errors across several popular benchmarks not only identifies the presence of errors but also quantifies their magnitude (e.g., error proportions in commonly used test/validation splits) and discusses the consequences for benchmark stability and model selection [27]. Even when rankings may in some cases remain robust, the normative problem persists: if the measurement basis itself is flawed, absolute performance values and fine distinctions—which frequently drive publication decisions—are less informative [27].

Finally, the proliferation of very large pretrained models (particularly large language models, LLMs) has introduced a new form of leakage: **benchmark contamination through pretraining data**. When benchmark items or semantically close paraphrases are contained in web-scale corpora, models can memorise portions of the evaluation data. Several studies develop detection methods, taxonomise contamination types, and propose documentation artefacts (e.g., "Benchmark Transparency Cards") to improve training-data transparency and comparability [28–30].

3 Limitations of Benchmarks and Optimisation Dynamics

The methodological problems identified in the previous section do not exist in isolation; they interact with the *institutional* structure of how AI research is evaluated. Benchmarks simultaneously serve as measurement instruments and as steering mechanisms for research attention and resource allocation. This dual role creates optimisation dynamics that can undermine the very validity benchmarks are designed to ensure.

3.1 Benchmark Optimisation as an Adaptive Inference Problem

In this respect, benchmarks resemble adaptive data analysis: hypotheses, models, and pipeline decisions are iteratively generated based on benchmark feedback. The statistics literature shows that repeated, adaptive use of a holdout dataset can undermine the validity of classical inference assumptions, because the "holdout" becomes implicitly integrated into the decision space [31].

In ML competitions and leaderboards this problem is operationally particularly visible: participants can overfit their solutions to a public leaderboard through repeated submission. Theoretical work models this risk and proposes mechanisms designed to reduce leaderboard

overfitting by disclosing holdout information only in a restricted, adaptation-resilient manner [32]. This perspective transfers to scientific benchmarks, where iterative tuning and "leaderboard chasing" are typical research practices, especially when improvements on the order of a few tenths of a percentage point suffice for publication [6].

3.2 Goodhart Effects, "General" Benchmarks, and External Validity

When a metric becomes a target, it can lose its informational value (Goodhart effect). In AI, this manifests in several forms: *(i)* models learn spurious correlations and shortcuts instead of intended capabilities; *(ii)* task metrics can be systematically "gamed" without improving semantic quality; *(iii)* benchmark design can unintentionally steer development toward metrically favourable but scientifically or socially less valid properties. Specific analyses demonstrate Goodhart mechanisms even within specialised benchmark families (e.g., explanation benchmarks), where scores can be substantially increased without genuine improvement in explanatory substance [10].

The critique of "general" benchmarks goes further, arguing that a small canon of prominent datasets in various subfields can be misinterpreted as a proxy for "general progress." A positional analysis describes such general benchmarks as structurally incompatible with strong claims of general capability, because benchmarks are necessarily under-specified and the reinterpretation of state-of-the-art (SOTA) performance on a benchmark as "general intelligence" is epistemically unjustified [11].

The external-validity conflict is further exacerbated by **dataset shift**: when the data distribution differs between training and test sets—or between benchmark and deployment—benchmark performance can only limitedly predict real-world performance. The problem has been discussed for years under the headings "dataset shift" and "covariate shift," including consequences for evaluation design and robust modelling [33]. Empirical benchmark research replicates this insight with new test sets or "fresh samples": when models are adapted to a static test-set canon over years, performance drops on newly drawn, but "equivalently intended," test sets can be interpreted as indicators of benchmark overfitting [34].

3.3 Benchmark Design Beyond Leaderboards: Suites Instead of Single-Number Rankings

A growing strand of the literature argues for transitioning from leaderboards to **benchmark suites**: instead of optimising "one number" (and hence one ranking), different metrics, usage scenarios, and trade-offs should be made explicit. This is articulated particularly clearly in fairness evaluation: "fairness" is conceptually plural and application-dependent;

a leaderboard that reduces fairness to a single score produces target distortions and can obscure potential harms. Benchmark suites, by contrast, are intended to make different perspectives and measurement dimensions visible, thereby increasing both scientific validity and governance-related decision-making capacity [35].

4 Evidence from Meta-Research and Open Science Initiatives

Having established the theoretical mechanisms through which biases and benchmark dynamics compromise AI research, we now turn to the empirical evidence. Meta-research quantifies the extent of the problem, while Open Science initiatives represent the community's institutional response.

4.1 Meta-Research: Findings on Documentation and Reproduction Practice in AI

Meta-research in AI consistently shows that pure results reports frequently do not contain sufficient information to reliably reproduce experiments. A documentation-oriented analysis spanning several major AI conferences (sample across multiple years) indicates, for example, that only a small proportion of the examined papers share centrally relevant artefacts (including code, results/outputs, hypotheses/predictions) and that experimental details (dependencies, setup parameters) can be severely under-documented [9]. These findings should not be read as mere "compliance" criticism; they represent an *epistemic* risk: when essential pipeline decisions are not auditable, it is difficult to distinguish whether a result rests on a robust methodological innovation or on unreported, potentially data-specific or chance-driven factors [3].

Meta-research on variance and benchmarking additionally reveals that benchmark values are often reported as point estimates despite being subject to stochastic and pipeline-related fluctuations. Work on "variance in machine learning benchmarks" models the benchmark pipeline and argues that common comparison practices underestimate variance, thereby producing a distorted picture of methodological superiority [6]. For subfields such as reinforcement learning, this problem is formulated particularly sharply: non-determinism in environments and intrinsic variance can render purported SOTA improvements difficult to interpret without significance metrics and robust reporting standards [5].

Finally, meta-research shows that not only *results reproduction* but also *inferential reproducibility* (the validity of conclusions across groups of tasks/datasets) constitutes a distinct target criterion: rather than "determinising variance away" (e.g., by fixing a single seed), the proposal is to deliberately vary variance sources and to model them statistically

(e.g., mixed-effects models) in order to obtain generalisable statements [19].

4.2 Open Science Initiatives: Checklists, Artifact Reviews, and Community Programmes

At the institutional level, several initiatives have emerged in recent years to raise reproducibility in AI. A central example is the introduction of **paper checklists** by major ML conferences, which query best practices regarding reproducibility, transparency, ethics, and societal impact; the guidelines emphasise that the checklist must not be removed and functions as a component of the submission [2].

An empirically well-documented intervention case is a conference **reproducibility programme** that combines a code-submission policy, a reproducibility challenge, and a checklist. The programme report describes, among other things, a marked increase in voluntary code submissions (from below 50% to nearly 75% within one year) and further reports that a portion of reviewers rated the checklist as helpful. At the same time, it emphasises that robust evidence on long-term quality effects remains limited, underscoring the need for further meta-evaluations of such interventions [1].

Beyond conference mechanisms, **artifact review and badging programmes** (particularly in the computing ecosystem) define criteria under which artefacts are evaluated as repeatable/reproducible and how this is made visible. Such frameworks aim to close the "artefact gap" between paper and executable research and to create incentives for clean packaging and documentation practices [36].

Open Science is further supported by broad, cross-disciplinary principles and manifestos that formulate transparency, openness, and reproducibility as systemic goals. Key reference points include *(i)* recommendations for promoting an open research culture [13], *(ii)* guidelines for the reproducibility of computational methods, particularly code and data provision [7, 8], and *(iii)* FAIR principles, which emphasise the machine-actionability of research objects [12].

5 Technical and Organisational Solutions

The preceding sections have established that the reproducibility crisis in AI arises from an interplay of statistical, benchmark-related, and documentation deficits. Addressing these deficits requires not only cultural and procedural change but also concrete technical infrastructure and governance mechanisms. This section reviews solutions along three axes: versioning and executable reproduction, dataset governance, and automated verification.

5.1 Versioning, Provenance, and Executable Reproduction

Technical reproducibility in AI requires the controlled management of code, data, model artefacts, and execution environments. Data and code versioning systems aim to make states (snapshots) of data and pipelines, as well as their dependencies, traceable.

One example is **DataLad**, a distributed system for the joint management of code and data that connects Git-based workflows with large-scale data logistics and is explicitly oriented toward reproducible research and FAIR compatibility [37]. For Data Version Control (DVC), empirical analyses of usage in open-source ML development characterise DVC as a tool for versioning and reproducibility of ML artefacts [38].

A second core building block is the **long-term archiving and referenceability of software sources**. Initiatives for comprehensive source-code archiving argue that persistent identifiers and systematic archiving can address central reproducibility barriers by reducing "link rot" and the problem of untraceable code states [39].

Third, **packaging tools and containerised reproduction environments** become relevant: reproducibility frequently fails due to "dependency hell." Tools for packaging experiments extract dependencies and enable cross-platform re-execution; corresponding systems have been positioned at data-management and database conferences as reproducibility infrastructure [40, 41]. Complementary platform approaches bundle code, data, dependencies, and metadata in a "capsule," check executability, and assign DOIs for citability and archiving; this is described in the literature as a way to reduce review and reproduction burden [42].

5.2 Dataset Governance: Documentation, Quality Assurance, and Citation Practices

Many reproducibility problems are data-driven. Consequently, a research line has emerged that understands dataset and model documentation as a governance instrument.

Datasheets for Datasets propose standardised documents that make the motivation, composition, collection process, recommended use, and limitations of datasets transparent; in peer-reviewed form, such standards have been prominently published [43]. For NLP, **Data Statements** exist as a professional practice format that systematically describes data provenance, sampling, annotation, and potential bias consequences to enable better science and bias mitigation [44]. For models, comparable documentation formats (**Model Cards**) have been proposed that disclose performance profiles, deployment contexts, and limitations [45].

A leakage-specific extension of this governance idea takes the form of **Model Info Sheets** as a checklist format that addresses concrete leakage types and thereby supports the detection of methodological errors prior to publication. Empirical evidence from ML-based science suggests that such systematic audit frameworks are necessary because

leakage errors are frequently not discernible from paper text alone [14].

Citation practices constitute a further governance element because they govern reusability, attribution, and machine-linkability of research objects. For data, a Joint Declaration has been formulated that describes data citation as part of reproducible scholarship and calls for machine- and human-readable citations [46]. For software, software citation principles have been published in peer-reviewed form, defining specificity (version), persistence, and attribution as core elements [47]. In practice, these goals are supported by formats such as the Citation File Format (CFF), in which citable metadata are stored in a machine-readable manner [48].

Finally, **research packaging standards** (e.g., RO-Crate) address the problem that paper text is often incomplete or ambiguous for automated systems. RO-Crate is described as a lightweight, community-driven packaging format that bundles research artefacts and metadata in a machine-readable fashion, thereby supporting FAIR goals and reproducibility [49].

5.3 Automated Citation and Evidence Verification Systems

With the growth of scientific literature and the advent of AI-generated content, the question of whether citations actually support the assertions they accompany becomes increasingly pressing. The connection to the preceding infrastructure is direct: even the best versioning and documentation frameworks provide limited value if the textual claims in papers do not accurately reference the artefacts they are supposed to describe.

The NLP literature has formalised benchmarks and tasks for this purpose: **scientific claim verification** establishes datasets in which claims are linked to evidence texts and annotated as "supports/refutes," enabling the evaluation of systems for evidence retrieval and reasoning [50]. Complementary subtasks such as **cite-worthiness detection** (where is a citation needed?) and **citation intent classification** have been developed, forming structural precursors to automated citation quality assurance [51].

More recent benchmarks explicitly evaluate whether language models can correctly attribute scientific claims and discuss the question of whether systems can reliably perform the mapping "claim → source." Such work positions automated attribution both as a means against hallucinations and as a building block for future verification systems [52]. At the system level, proposals exist for **citation verification** through full-text analysis with classification taxonomies (e.g., supported/unsupported) and provision of evidence snippets; corresponding systems argue that scalable citation checking is integrity-relevant, especially for peer review and AI-generated content [53].

For the reproducibility crisis in AI, this line of work is particularly relevant for two reasons. First, automated citation checking can *enforce* the referencing of concrete artefacts (dataset versions, code snapshots, model hashes) by detecting missing or incorrect references

and suggesting corrections. Second, it can serve as a quality-assurance mechanism in review pipelines to reduce the discrepancy between textual assertions and actual evidence/artefacts. Both roles are explicitly formulated as target directions in the literature—integrity, peer-review relief, and evidence-based justification [53].

6 Case Studies

The mechanisms and solutions discussed so far can be illustrated through concrete cases drawn from different AI subfields. Each case study highlights a different facet of the reproducibility crisis and connects back to the theoretical framework established in the preceding sections.

6.1 Reinforcement Learning: Variance, Non-Determinism, and Inference Uncertainty

An influential analysis of deep reinforcement learning problematises the fact that non-deterministic environments and methodological variance can render reported performance gains difficult to interpret when significance measures, multiple runs, and standardised reporting practices are lacking. The finding is not merely an engineering problem; it is a *statistical inference problem*: without variance modelling, it remains unclear whether differences are robust or chance-driven [5]. This case exemplifies the p-hacking analogue discussed in section 2.1: researchers may select the best seed or environment configuration, generating an illusory impression of progress.

6.2 Computer Vision Benchmarks: Near-Duplicates and Measurement Foundations

The ciFAIR study demonstrates that near-duplicates between training and test sets exist in heavily used benchmarks and that a "purged" test set can reveal considerable performance losses. The study interprets this as an indication that benchmark progress can be partly facilitated by memorisation and that benchmark design and dataset hygiene are central components of scientific validity [26]. Complementary work on label errors in test sets shows that even test data cannot be presupposed as "ground truth" and that label quality can influence the stability of benchmark comparisons [27]. Together, these findings reinforce the leakage and data-quality concerns raised in section 2.2.

6.3 LLM Evaluation: Contamination and Dynamic Benchmarks

Several studies document that benchmark contamination—overlap between evaluation data and pretraining corpora—can inflate performance claims and undermine fair comparisons;

proposed remedies include detection pipelines, survey systematisations, and transparency artefacts [28, 29]. At the same time, more recent analyses show that the effect of contamination on generative benchmarks is causally modulated by parameters such as sampling temperature and solution/response length; this means that "contamination" must be modelled not as a binary problem but as a context-dependent source of bias [54].

In response, **contamination-limited benchmarks** have emerged that use frequently updated, objectively scorable tasks to reduce contamination risk and to increase the reliability of LLM evaluation [55]. This development represents a practical implementation of the benchmark-suite philosophy discussed in section 3.3.

6.4 ML-Based Science: Leakage as a Reproduction Error

The leakage meta-study in ML-based science demonstrates that leakage is not merely an internal ML error class but can have reproducibility-critical consequences for entire application domains. In a reproduction study on civil-war prediction, it shows that the purported superiority of complex ML models disappears upon correction of leakage, thereby invalidating central conclusions of the original literature [14]. This case underscores the broader point made throughout this paper: performance claims that rest on compromised evaluation pipelines can propagate unchecked through the scientific record if reproducibility infrastructure and governance are absent.

7 Discussion

The evidence assembled in this paper paints a consistent picture: the "reproducibility crisis" in AI is neither exclusively a problem of "missing code links" nor solely a problem of "poor scientific culture." Rather, it emerges from the interplay of statistical selection logic, benchmark incentive mechanisms, and the infrastructural complexity of computational experiments [3, 14].

A central insight is that AI-typical degrees of freedom can assume the role of multiple hypothesis tests: configurations are iteratively optimised, and the final report often condenses this process into a single score. The methodological consequence is that scientific standards must be oriented more strongly toward measurement uncertainties, variance components, and robust inference (e.g., multiple seeds, appropriate test protocols, effect sizes instead of "best score") [6, 19].

Benchmarks fulfil an economic function (standardisation, comparability), but when misused they can generate epistemic damage: adaptive overfitting, Goodhart effects, conflation of proxy targets with "capabilities," and erosion of external validity through dataset shift. This does not mean that benchmarks are obsolete; it means that their *governance*—suites instead of leaderboards, regular updating/rotation, leakage mitigation, quality control of

labels and duplicates—becomes a core component of scientific methodology [35].

The intervention literature suggests that community mechanisms can be effective in shifting norms (e.g., rising code submission rates, establishment of checklists), but the evidence on long-term effects on research quality remains incomplete. This implies a *meta-evaluation imperative*: reproducibility programmes should themselves be understood as evaluable interventions whose effectiveness (quality, robustness, generalisability) is empirically tracked [1].

Technically, a design principle emerges: reproducibility requires a chain of *referenceability* (persistent IDs), *versioning* (code/data/models), *packaging* (executable environments), *documentation* (Datasheets / Data Statements / Model Cards / Info Sheets), and finally *auditability* (including automated evidence and citation verification). Each link addresses a different error class: versioning reduces "invisible" drift; packaging reduces execution barriers; documentation reduces interpretation and governance deficits; automated citation checking reduces "paper-to-artefact" inconsistencies and raises the probability that claims actually rest on traceable evidence [37, 49].

8 Conclusion

The reproducibility crisis in AI is best understood as a *multi-layered validity problem*:

(i) statistical biases through selection and reporting mechanisms;

(ii) methodological error classes such as leakage, overfitting, duplicates, and erroneous labels;

(iii) a benchmark economy with adaptive overuse and Goodhart effects; and

(iv) infrastructural fragility through complex software and data dependencies [14].

An effective solution approach must therefore simultaneously:

(a) **modernise statistical inference practices**—variance modelling, robust tests, transparent reporting standards;

(b) **reform benchmark governance**—suites, updating, leakage control;

(c) **strengthen Open Science institutions**—checklists, artifact review, replication publication; and

(d) **expand technical infrastructure** toward versioned, packaged, and citable research objects—complemented by automated evidence and citation verification as a scaling mechanism for quality assurance [1].

None of these measures is sufficient in isolation. Statistical reform without infrastructure leaves researchers unable to share artefacts even when willing; infrastructure without governance risks creating technically reproducible but inferentially flawed research; governance without statistical discipline cannot prevent the selection biases that arise in research logic itself. Only their *combination*—a coherent framework spanning statistics, benchmarks, institutions, and technology—can credibly address the multi-dimensional nature of the crisis and rebuild confidence in the empirical foundations of artificial intelligence.

References

[1] Joelle Pineau, Philippe Vincent-Lamarre, Koustuv Sinha, Vincent Larivière, Alina Beygelzimer, Florence d'Alché Buc, Emily Fox, and Hugo Larochelle. Improving reproducibility in machine learning research: A report from the NeurIPS 2019 reproducibility program. *Journal of Machine Learning Research*, 22(164):1–20, 2021. URL https://jmlr.org/papers/v22/20-303.html.

[2] NeurIPS. NeurIPS paper checklist guidelines, 2023. URL https://neurips.cc/public/guides/PaperChecklist. Accessed: 2025.

[3] Steven N. Goodman, Daniele Fanelli, and John P. A. Ioannidis. What does research reproducibility mean? *Science Translational Medicine*, 8(341):341ps12, 2016. doi: 10.1126/scitranslmed.aaf5027.

[4] National Academies of Sciences, Engineering, and Medicine. Reproducibility and replicability in science. Technical report, The National Academies Press, Washington, DC, 2019.

[5] Peter Henderson, Riashat Islam, Philip Bachman, Joelle Pineau, Doina Precup, and David Meger. Deep reinforcement learning that matters. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2018. doi: 10.5555/3504035.3504427. URL https://arxiv.org/abs/1709.06560.

[6] Xavier Bouthillier, Pierre Delaunay, Mirko Bronzi, Assya Trofimov, Brennan Nichyporuk, Justin Szeto, et al. Accounting for variance in machine learning benchmarks. In *Proceedings of Machine Learning and Systems (MLSys)*, volume 3, 2021. URL https://proceedings.mlsys.org/paper_files/paper/2021/hash/0184b0cd3cfb185989f858a1d9f5c1eb-Abstract.html.

[7] Roger D. Peng. Reproducible research in computational science. *Science*, 334(6060): 1226–1227, 2011. doi: 10.1126/science.1213847.

[8] Victoria Stodden, Marcia McNutt, David H. Bailey, Ewa Deelman, Yolanda Gil, Brooks Hanson, Michael A. Heroux, John P. A. Ioannidis, and Michela Taufer. Enhancing reproducibility for computational methods. *Science*, 354(6317):1240–1241, 2016. doi: 10.1126/science.aah6168.

[9] Odd Erik Gundersen. On reproducible AI. Technical report, Norwegian University of Science and Technology, 2020. URL https://folk.idi.ntnu.no/odderik/RAI-2020/On_Reproducible_AI-preprint.pdf.

[10] Jennifer Hsia, Danish Pruthi, Aarti Singh, and Zachary C. Lipton. Goodhart's law applies to NLP's explanation benchmarks. In *Findings of the Association for*

Computational Linguistics: EACL 2024, 2024. URL https://aclanthology.org/2024.findings-eacl.88/.

[11] Inioluwa Deborah Raji, Emily M. Bender, Amandalynne Paullada, Emily Denton, and Alex Hanna. AI and the everything in the whole wide world benchmark. In *Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks*, 2021. URL https://openreview.net/forum?id=j6NxpQbREA1.

[12] Mark D. Wilkinson, Michel Dumontier, IJsbrand Jan Aalbersberg, Gabrielle Appleton, Myles Axton, Arie Baak, et al. The FAIR guiding principles for scientific data management and stewardship. *Scientific Data*, 3:160018, 2016. doi: 10.1038/sdata.2016.18.

[13] Brian A. Nosek, George Alter, George C. Banks, Denny Borsboom, Sara D. Bowman, Steven J. Breckler, et al. Promoting an open research culture. *Science*, 348(6242): 1422–1425, 2015. doi: 10.1126/science.aab2374.

[14] Sayash Kapoor and Arvind Narayanan. Leakage and the reproducibility crisis in machine-learning-based science. *Patterns*, 4(9):100804, 2023. doi: 10.1016/j.patter.2023.100804.

[15] Marcus R. Munafò, Brian A. Nosek, Dorothy V. M. Bishop, Katherine S. Button, Christopher D. Chambers, Nathalie Percie du Sert, Uri Simonsohn, Eric-Jan Wagenmakers, Jennifer J. Ware, and John P. A. Ioannidis. A manifesto for reproducible science. *Nature Human Behaviour*, 1:0021, 2017. doi: 10.1038/s41562-016-0021.

[16] Joseph P. Simmons, Leif D. Nelson, and Uri Simonsohn. False-positive psychology: Undisclosed flexibility in data collection and analysis allows presenting anything as significant. *Psychological Science*, 22(11):1359–1366, 2011. doi: 10.1177/0956797611417632.

[17] John P. A. Ioannidis. Why most published research findings are false. *PLoS Medicine*, 2(8):e124, 2005. doi: 10.1371/journal.pmed.0020124.

[18] Megan L. Head, Luke Holman, Rob Lanfear, Andrew T. Kahn, and Michael D. Jennions. The extent and consequences of p-hacking in science. *PLoS Biology*, 13(3): e1002106, 2015. doi: 10.1371/journal.pbio.1002106.

[19] Michael Hagmann et al. Towards inferential reproducibility of machine learning research. *arXiv preprint*, 2023. doi: 10.48550/arXiv.2302.04054. URL https://arxiv.org/abs/2302.04054.

[20] Janez Demšar. Statistical comparisons of classifiers over multiple data sets. *Journal of Machine Learning Research*, 7:1–30, 2006. URL https://jmlr.org/papers/v7/demsar06a.html.

[21] Rotem Dror, Gili Baumer, Segev Shlomov, and Roi Reichart. The hitchhiker's guide to testing statistical significance in natural language processing. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 1383–1392, 2018. doi: 10.18653/v1/P18-1128.

[22] Annie Franco, Neil Malhotra, and Gabor Simonovits. Publication bias in the social sciences: Unlocking the file drawer. *Science*, 345(6203):1502–1505, 2014. doi: 10.1126/science.1255484.

[23] Daniele Fanelli. "positive" results increase down the hierarchy of the sciences. *PLoS ONE*, 5(4):e10068, 2010. doi: 10.1371/journal.pone.0010068.

[24] Christopher D. Chambers. Registered reports: A new publishing initiative at *Cortex*. *Cortex*, 49(3):609–610, 2013. doi: 10.1016/j.cortex.2012.12.016.

[25] Iulian Emil Tampu, Anders Eklund, and Neda Haj-Hosseini. Inflation of test accuracy due to data leakage in deep learning-based classification of OCT images. *Scientific Data*, 9:580, 2022. doi: 10.1038/s41597-022-01618-6.

[26] Björn Barz and Joachim Denzler. Do we train on test data? Purging CIFAR of near-duplicates. *Journal of Imaging*, 6(6):41, 2020. doi: 10.3390/jimaging6060041.

[27] Curtis G. Northcutt, Anish Athalye, and Jonas Mueller. Pervasive label errors in test sets destabilize machine learning benchmarks. In *Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks*, 2021. doi: 10.48550/arXiv.2103.14749.

[28] Ruijie Xu, Zengzhi Wang, Run-Ze Fan, and Pengfei Liu. Benchmarking benchmark leakage in large language models. *arXiv preprint*, 2024. doi: 10.48550/arXiv.2404.18824.

[29] Cheng Xu, Shuhao Guan, Derek Greene, and M.-Tahar Kechadi. Benchmark data contamination of large language models: A survey. *arXiv preprint*, 2024. doi: 10.48550/arXiv.2406.04244.

[30] Hyeong Kyu Choi et al. How contaminated is your benchmark? Measuring dataset leakage in LLMs with kernel divergence. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2025. URL https://proceedings.mlr.press/v267/choi25b.html.

[31] Cynthia Dwork, Vitaly Feldman, Moritz Hardt, Toniann Pitassi, Omer Reingold, and Aaron Roth. Preserving statistical validity in adaptive data analysis. In *Proceedings of the 47th Annual ACM Symposium on Theory of Computing (STOC)*, 2015. doi: 10.48550/arXiv.1411.2664. Originally arXiv 2014.

[32] Avrim Blum and Moritz Hardt. The ladder: A reliable leaderboard for machine learning competitions. In *Proceedings of the 32nd International Conference on Machine Learning (ICML)*, 2015. doi: 10.48550/arXiv.1502.04585.

[33] Joaquin Quiñonero-Candela, Masashi Sugiyama, Anton Schwaighofer, and Neil D. Lawrence. *Dataset Shift in Machine Learning*. MIT Press, 2009. URL https://mitpress.mit.edu/9780262545877/dataset-shift-in-machine-learning/.

[34] Benjamin Recht, Rebecca Roelofs, Ludwig Schmidt, and Vaishaal Shankar. Do ImageNet classifiers generalize to ImageNet? 2019. doi: 10.48550/arXiv.1902.10811. URL https://arxiv.org/abs/1902.10811.

[35] Angelina Wang, Aaron Hertzmann, and Olga Russakovsky. Benchmark suites instead of leaderboards for evaluating AI fairness. *Patterns*, 5(11):101080, 2024. doi: 10.1016/j.patter.2024.101080.

[36] ACM. Artifact review and badging, 2020. URL https://www.acm.org/publications/policies/artifact-review-badging. Version 1.1.

[37] Yaroslav O. Halchenko, Kyle Meyer, Benjamin Poldrack, Debanjum Singh Solanky, Adina S. Wagner, Jason Gors, et al. DataLad: Distributed system for joint management of code, data, and their relationship. *Journal of Open Source Software*, 6(63): 3262, 2021. doi: 10.21105/joss.03262.

[38] Lívia Barreto Simedo Pacheco et al. DVC in open source ML development: An empirical study. In *Proceedings of the ACM/IEEE International Conference on Mining Software Repositories (MSR)*, 2024. doi: 10.1145/3644815.3644965.

[39] Roberto Di Cosmo and Stefano Zacchiroli. Software heritage: Collecting, preserving, and sharing all software source code. In *Proceedings of the ACM/IEEE International Conference on Automated Software Engineering (ASE)*, 2018. doi: 10.1145/3238147.3241985.

[40] Fernando Chirigati, Rémi Rampin, Dennis Shasha, and Juliana Freire. ReproZip: Computational reproducibility with ease. In *Proceedings of the ACM SIGMOD International Conference on Management of Data*, pages 2085–2088, 2016. doi: 10.1145/2882903.2899401.

[41] Rémi Rampin, Fernando Chirigati, Dennis Shasha, Juliana Freire, and Vicky Steeves. ReproZip: The reproducibility packer. *Journal of Open Source Software*, 3(27):107, 2018. doi: 10.21105/joss.00107.

[42] Barbara Cheifet. Promoting reproducibility with Code Ocean. *Genome Biology*, 22: 65, 2021. doi: 10.1186/s13059-021-02299-x.

[43] Timnit Gebru, Jamie Morgenstern, Briana Vecchione, Jennifer Wortman Vaughan, Hanna Wallach, Hal Daumé III, and Kate Crawford. Datasheets for datasets. *Communications of the ACM*, 64(12):86–92, 2021. doi: 10.1145/3458723.

[44] Emily M. Bender and Batya Friedman. Data statements for natural language processing: Toward mitigating system bias and enabling better science. *Transactions of the Association for Computational Linguistics*, 6:587–604, 2018. doi: 10.1162/tacl_a_00041.

[45] Margaret Mitchell, Simone Wu, Andrew Zaldivar, Parker Barnes, Lucy Vasserman, Ben Hutchinson, Elena Spitzer, Inioluwa Deborah Raji, and Timnit Gebru. Model cards for model reporting. In *Proceedings of the Conference on Fairness, Accountability, and Transparency (FAT*)*, pages 220–229, 2019. doi: 10.48550/arXiv.1810.03993.

[46] Data Citation Synthesis Group. Joint declaration of data citation principles, 2014.

[47] Arfon M. Smith, Daniel S. Katz, and Kyle E. Niemeyer. Software citation principles. *PeerJ Computer Science*, 2:e86, 2016. doi: 10.7717/peerj-cs.86.

[48] Stephan Druskat, Jurriaan H. Spaaks, Neil Chue Hong, Robert Haines, James Baker, et al. Citation file format (CFF), 2021. URL https://citation-file-format.github.io/.

[49] Stian Soiland-Reyes, Peter Sefton, Mercè Crosas, Leyla Jael Castro, Frederik Coppens, José M. Fernández, Daniel Garijo, et al. Packaging research artefacts with RO-Crate. *Data Science*, 5(2):97–138, 2022. doi: 10.3233/DS-210053.

[50] David Wadden, Shanchuan Lin, Kyle Lo, Lucy Lu Wang, Madeleine van Zuylen, Arman Cohan, and Hannaneh Hajishirzi. Fact or fiction: Verifying scientific claims. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 7534–7550, 2020. doi: 10.18653/v1/2020.emnlp-main.609.

[51] Dustin Wright and Isabelle Augenstein. CiteWorth: Cite-worthiness detection for improved scientific document understanding. In *Findings of the Association for Computational Linguistics: ACL 2021*, 2021. doi: 10.48550/arXiv.2105.10912.

[52] Ofir Press, Avi Coplien, Ori Yoran, Yoav Goldberg, and Jonathan Berant. CiteME: Can language models accurately cite scientific claims? In *Proceedings of the Neural*

Information Processing Systems Conference (NeurIPS), Datasets and Benchmarks Track, 2024. URL https://proceedings.neurips.cc/paper_files/paper/2024/file/0ef47f7b768e1a012e3d995ac8d8fac7-Paper-Datasets_and_Benchmarks_Track.pdf.

[53] SemanticCite Team. SemanticCite: Citation verification with AI-powered full-text analysis. *arXiv preprint*, 2025. doi: 10.48550/arXiv.2511.16198. URL https://arxiv.org/abs/2511.16198.

[54] Rylan Schaeffer et al. Causally quantifying the effect of test set contamination on generative benchmarks. In *Proceedings of the Neural Information Processing Systems Conference (NeurIPS)*, 2025. URL https://neurips.cc/virtual/2025/122430.

[55] Harry White, Adam Mahdi, et al. LiveBench: A challenging, contamination-limited LLM benchmark. In *Proceedings of the International Conference on Machine Learning (ICML)*, 2024. URL https://openreview.net/forum?id=sKYHBTAxVa.